

Interactions of Gas Particles with Graphene during High-Throughput Compressible Flow Exfoliation: A Molecular Dynamics Simulations Study

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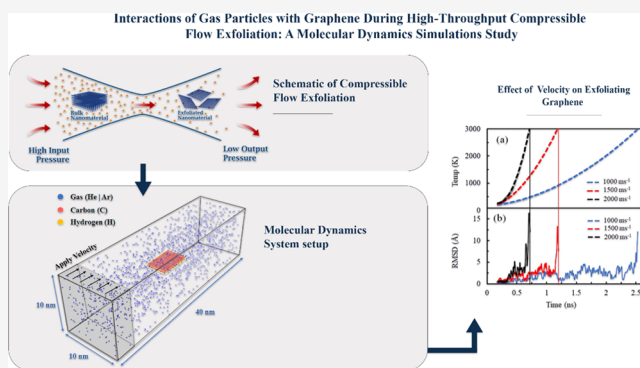
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ABSTRACT: The field of two-dimensional (2D) nanomaterials has gained significant interest over the last few decades in numerous applications because of their unique properties that exhibit when a bulk material is reduced to its 2D form. A wide variety of 2D layered materials are synthesized by a newly developed compressible flow exfoliation (CFE) process, which has considerable advantages over current top-down approaches. In this study, classical molecular dynamics (MD) simulations are used to investigate the interactions of gas particles with pristine, unfunctionalized graphene sheets during the CFE process and try to understand the atomistic mechanism of layer separation. The thermal vibration of graphene layers increases with the elevation of temperature that accelerates the exfoliation tendency, but the presence of static gas particles is insignificant here because of their lower binding energy. The range of one-directional flow velocities is incorporated to the compressible gases to replicate the experimental situation, and dispersion of graphene is observed when the velocity exceeds the supersonic flow condition. Analyzing the dynamic properties of exfoliation, it is established that sliding or the parallel direction is the preferable exfoliation mechanism of graphene than vertical separation. Besides, the upstream pressure plays a fundamental role because gas density and flow velocity are associated with that. It is also observed that heavier gas is less susceptible to delaminate graphene than lighter gas because of their higher atomic mass and lower flow rate at identical conditions. The findings of this study provide more flexibility to synthesize not only graphene but any 2D materials using compressible gases.



INTRODUCTION

The wonder material graphene, a one-atom-thick sheet of carbon atoms arranged in a honeycomb structure, is considered as the world's thinnest, strongest, and most conductive material. It has attracted a great deal of attention since its discovery in 2004 because of its high specific surface area, as well as outstanding mechanical, electrical, thermal, and optical properties,^{1–3} and they can be used in key electronics, sensors, hydrogen and energy storage,⁴ water desalination,⁵ catalysis,⁶ composite materials,⁷ and biomedicine.⁸ The properties of graphene are mostly dependent on their lattice structure, thickness, types, and number of defects present in the materials.⁹ Until now, a variety of processes have been developed to synthesize graphene, but only a few of them are efficient and cost effective to produce high-quality and less-defective graphene.

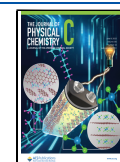
Currently, graphene nanosheets are obtained either by chemical synthesis or physical separation or a combination of both. These fabrication processes can be categorized into two major groups: bottom-up and top-down approaches.^{10,11} In the first approach, graphene grows on a suitable substrate from

various organic precursors using the physical or chemical vapor deposition (CVD) method.¹² This method is suitable for synthesizing a large-area graphene sheet but it is difficult to transfer ultrathin graphene onto substrates. Conversely, exfoliation of graphene layers has been achieved by overcoming van der Waals interactions between layers of graphite in the top-down approach. Mechanical cleavage is a simple top-down method for high-quality graphene,¹³ but unfortunately, it is a very slow approach and always remains a great challenge to synthesize a large-quantity graphene. Chemical approaches like liquid-phase exfoliation (LPE) are widely considered to be a promising approach for the large-scale production of graphene, which is based on the dispersion and exfoliation of graphene oxide or graphite intercalation compounds (GIC).¹⁴ However,

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graphene synthesized by this process contains a significant amount of oxygen and other functional groups with a number of defects that cannot be restored completely by long and complicated postprocessing.¹⁵

Recently, exfoliation of two-dimensional (2D) nanomaterials using supercritical fluid has also gained significant interest. Perhaps, Pu et al. first demonstrated the ability of sCO₂ to overcome the van der Waals attraction force between graphite layers to synthesize graphene. Carbon dioxide (CO₂),¹⁶ ethanol,¹⁷ and *N,N*-dimethylformamide¹⁸ are most commonly used SCF media right now. The effect of temperature, pressure, shock velocity,¹⁹ sCO₂ density,²⁰ colloid aggregation,²¹ aqueous solution content ratio, and ultrasonic power²² during the exfoliation of graphene has been investigated in many studies. However, apart from supercritical fluids, the potential of other gaseous media to exfoliate 2D materials has not been explored that much. Our group has evolved a new top-down approach to synthesize 2D-layered materials using the supersonic flow of gases, which is titled as compressible flow exfoliation (CFE).²³ Continuous flow of high-pressure gas carries the bulk materials through a small orifice or a C-D nozzle, and the expanded gas–solid mixture is sprayed into a suitable solvent (Figure 1). A maximum concentration of 0.2

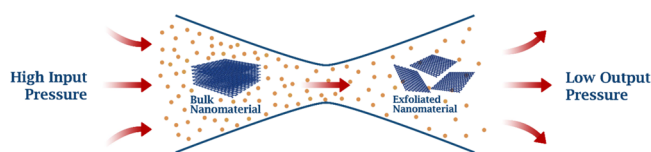


Figure 1. Schematic of compressible flow exfoliation of 2D layered materials.

mg mL^{−1} has been achieved with a mean thickness of 4.2 nm and a mean length of 276 nm (aspect ratio ≈65) in case of h-BN. In comparison to other liquid shear-based and supersonic fluid processes, it is not time-dependent and is very fast, which causes less defects in the finish products and environmentally friendly because there is no toxic chemical involves in postprocessing. Nanocomposites produced using CFE-processed 2D nanomaterials have achieved high mechanical and barrier properties that can create a lot of applications in plastic industry. The biggest advantage of the CFE process is its

versatility to be applied to any 2D layered materials using any gaseous substance.

Molecular dynamics (MD) simulation can be considered as computational experiments to explore physical movements of atoms and molecules that are beyond the accessibility of modern experimental approaches. It can also be referred to as statistical mechanics by numbers and Laplace's vision of Newtonian mechanics to predict the interatomic interactions by mimicking natural forces.²⁴ The continual increase of computing power will make it possible to replicate the experimental scenario in real timescale in near future. Classical MD has been successfully used to analyze the exfoliation phenomena of 2D nanomaterials in different synthesis processes like LPE,²¹ ion intercalation,²⁵ micromechanical cleavage,²⁶ electrochemical exfoliation,²⁷ and supercritical fluid.²⁸ In those studies, the simulations have been divided into two major categories: intercalation and exfoliation. First, it has been tried to determine the minimum separation distance between layers of 2D nanosheets, which will be necessary for solvent or fluid particles to diffuse. Second, the system of 2D nanomaterials has been modeled with that bilayer separation distance for exfoliation simulation. It was claimed that sonication or mechanical agitation can create this gap between layers, which is very unlikely. This artificial increase in the separation distance reduces the van der Waals interaction energy between layers, which makes it easier for fluids to intercalate and exfoliate nanosheets. However, the result of previous studies might be altered if the bulk materials are allowed to separate from their standard natural structure.

In the present work, we have used large-scale molecular dynamics simulation to investigate the microscopic mechanism behind the exfoliation behavior of the graphene bilayer in the presence of gas particles during compressible flow exfoliation (CFE). The findings of this study provide some useful information that may allow to maximize yield through the design of experiments and increase the versatility of this process. To understand the effect of temperature in the diffusion of graphene, two types of simulation have been done: (1) without the presence of gas particles and (2) with helium gas particles. Next, we have determined the optimum flow velocity of helium that is required for graphene separation based on their root-mean-square-deviation (RMSD) value, which is further justified through the self-diffusion coefficient of delaminating graphene. Apart from that, analyzing the

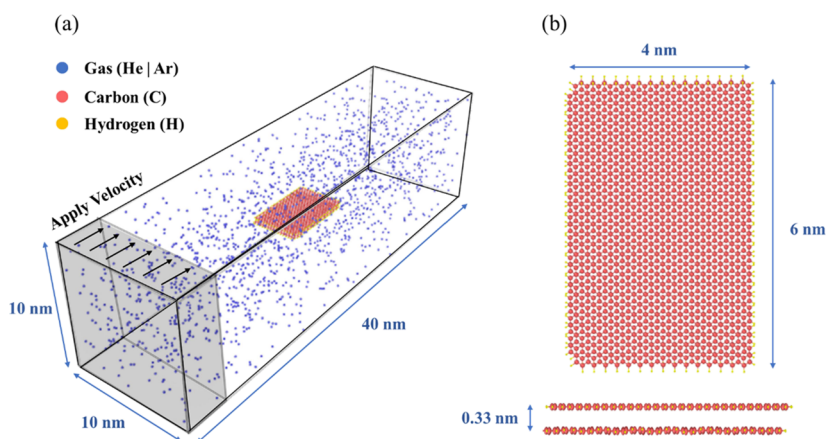


Figure 2. (a) System setup, a 10 × 40 × 10 nm³ box is considered with 5 nm entrance to apply velocity to gas particles (b) top view and side view of the graphene nanosheet of 6 × 4 nm² with 0.33 nm interlayer separation distance.

dynamic properties of exfoliation on molecular level, we have compared the possibilities of sliding and vertical separation pathways of 2D nanosheets. Based on experimental data, we have also studied the effect of upstream gas pressure on the diffusion phenomena of graphene because gas density and flow velocity are intimately related to the process pressure. Furthermore, heavier gas particles like argon were used to compare results with helium, and our investigation has established that the atomic mass of gas has a significant effect on graphene exfoliation. Unlike simulations of other exfoliation approaches, we construct the graphene structure with a standard bilayer distance (3.35 Å) during all exfoliation simulation: a stringent requirement for demonstrating layer separation. Combined with experimental and numerical estimates of gas density and flow velocity, this combined theoretical framework can provide fundamental insights into the atomistic level for enhancing dispersion of graphene as well as other 2D nanomaterials in any gaseous medium.

COMPUTATIONAL DETAILS

In the present study, large-scale atomic/molecular massively parallel simulator (LAMMPS) package is used for performing molecular dynamics calculations.²⁹ As the accuracy of the estimated properties by MD simulation mainly depends on the precision of potential functions used, we have employed the adaptive intermolecular reactive empirical bond order–Morse (AIREBO-M)³⁰ potential parameters, which is more suitable for the high-pressure simulation of graphene systems.

We first studied the effect of temperature on the exfoliation behavior of graphene nanosheets. In this regard, two approaches were adapted at static conditions: simulation of the graphene bilayer, (1) without any gaseous medium and (2) with helium gas particles. For both cases, we have constructed two armchair graphene nanosheets each having the dimension of $6 \times 4 \text{ nm}^2$ and separated by 0.33 nm, as shown in Figure 2. The bilayer graphene was placed at the center of a simulation box of dimension $10 \times 40 \times 10 \text{ nm}^3$. Afterward, a range of temperature (300–2000 K) was applied to the system using an NVT ensemble.

Next, we incorporated different flow velocities to the gaseous particles to observe the exfoliation phenomena of the static graphene bilayer. The system was equilibrated in the NVT ensemble at 300 K, which was followed by production run in the NVE ensemble. The flux of gas was reached to the required constant velocity through a stepwise method. To keep velocity at a constant value, the velocity of gas atoms inside the $10 \times 10 \times 10 \text{ nm}^3$ box at the entrance of the simulation box (Figure 2a) was monitored during the simulation, and every 2 ps the difference between the desired velocity and current velocity was added to gas atoms. In all simulations, one of the graphene sheets was harmonically constrained with a force constant of 185 kcal/mol/Å^2 during the production period to monitor the exfoliation of other graphene sheets, which was allowed to move freely. Hence, the fixed graphene sheet was considered as a terminal sheet of the bulk graphite.²¹

The equations of motion were integrated with a time step of 0.5 fs using the velocity Verlet Algorithm,³¹ and the nonbonded interactions were calculated setting a cutoff distance of 0.75 nm. A Noose–Hoover³² thermostat and a barostat were implemented to control the temperature and pressure of the system, respectively. The periodic boundary condition was applied in x and y directions, and the z direction was remained fixed.

AIREBO force field is widely used for the hydrocarbon system to determine their dynamic bonding process at ambient conditions.³³ Unfortunately, this energy function fails when the system is subjected to higher pressure. Therefore, the AIREBO-M³⁰ force field was used in this study, where diverging singular Lennard-Jones potential is replaced by a nondivergent Morse potential. It has an additional parameter that allows the repulsive region to be softened with little change to the minimum and attractive regions. The equation of the potential energy function used in the simulation has the form

$$E = \frac{1}{2} \sum_i \sum_{j \neq i} \left[E_{ij}^{\text{REBO}} + E_{ij}^{\text{LJ}} + \sum_{k \neq i,j} \sum_{l \neq i,j,k} E_{ijkl}^{\text{TORSION}} \right] \quad (1)$$

There are three terms in this energy function. The E_{REBO} term in the AIREBO-M potential gives the model its reactive capabilities and only describes short-ranged C–C, C–H, and H–H interactions ($r < 2 \text{ Å}$). The E_{LJ} term adds longer-ranged interactions ($2 < r < \text{cutoff}$) using a form like the standard 12–6 Lennard-Jones potential, but this fails at many pressure systems. To improve the accuracy of the function in such situation, the LJ potentials were replaced with Morse potential:³⁰

$$E_{ij}(r) = -\epsilon_{ij} [1 - (1 - e^{-\alpha(r-r_{ij}^{\text{eq}})})^2] \quad (2)$$

where ϵ and r_{ij}^{eq} define the depth and location of the minimum energy, respectively, and the new parameter α modifies the curvature of the potential energy at its minimum separation. This extra degree of freedom in the Morse potential is fit to experimental measurements of the layer spacing of graphite up to 14 GPa while retaining most equilibrium properties of AIREBO.³³ Also, the remaining E_{TORSION} term in the potential function is an explicit four-body potential that describes various dihedral angle preferences in hydrocarbon configurations.

Two different monoatomic gases are used in this study to investigate the interactions of gas particles with graphene during exfoliation. One is the lightest noble gas–helium and another one is its heavier counterparts–argon. Lennard-Jones (LJ) potential function is used here for mapping their interatomic interactions (Table 1).^{34,35} All nonbonded

Table 1. LJ Interaction Parameters^{34,35}

interaction	σ (Å)	ϵ (kcal/mol)
He–He	0.020	2.57
C–He	0.037	3.06
Ar–Ar	0.236	3.41
C–Ar	0.128	3.48

interactions are calculated from OPLS-AA (optimized potentials for liquid simulations–all atom) potential function.³⁶ Lorentz–Berthelot mixing rules³⁷ are used for approximating the interactions between unlike pairs like C–He, H–He, C–Ar, and H–Ar.

RESULTS AND DISCUSSION

To analyze the role of temperature on the exfoliation phenomena of bulk graphite, we considered two armchair graphene nanosheets separated by 0.33 nm, which is placed in

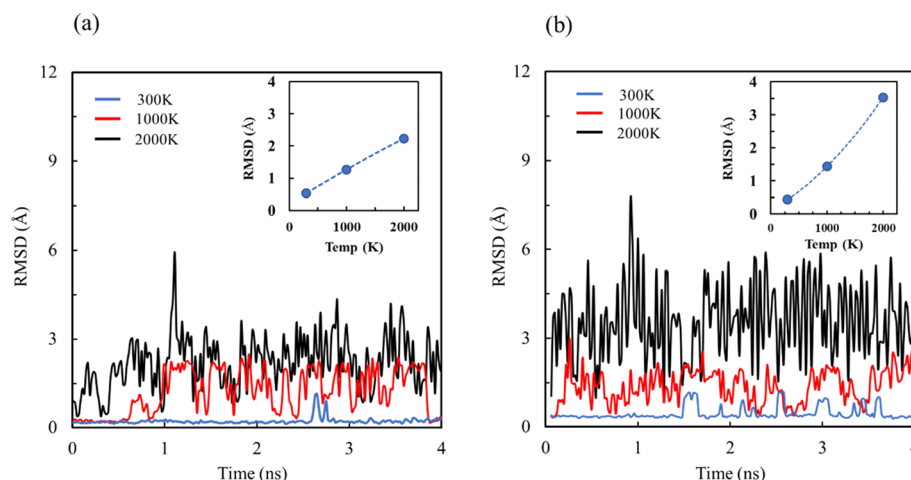


Figure 3. RMSD plot of the exfoliating graphene layer at five different temperatures ranging from 273 to 2000 K. (a) Graphene bilayer without any gaseous medium and (b) graphene bilayer with helium gas particles.

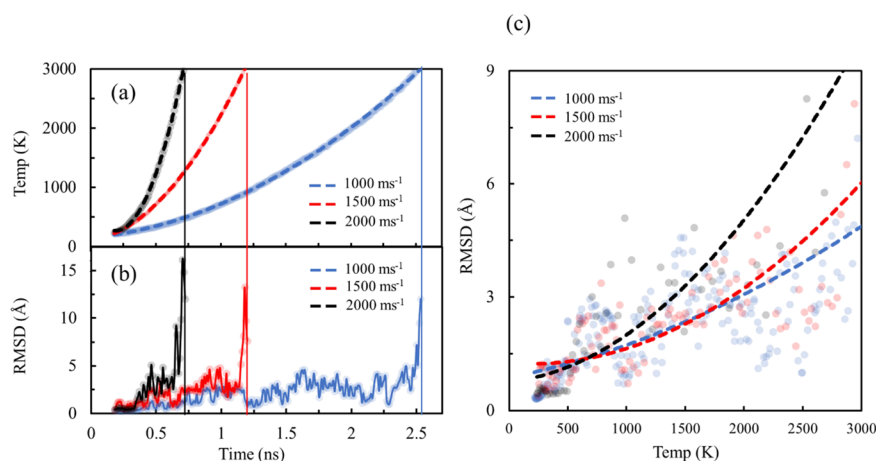


Figure 4. Effect of flow velocity of helium gas ranging from 1000 to 2000 ms^{-1} on graphene exfoliation (a) temperature and (b) RMSD of bilayer graphene. (c) Correlation of temperature and RMSD.

the center of an empty simulation box. As noted earlier, the bottom layer of the graphene is harmonically fixed to easily monitor the separation of another layer. It is always better to consider multilayers of graphene sheets to replicate the effects of bulk graphite, but the extended size of the system becomes computationally very expensive and makes the exfoliation intractable at longer timescales. That is why, a two-layer model of graphene nanosheets is considered here, which has been extensively used in various literature.^{38,39} Figure 3 shows the RMSD of the exfoliating layer of graphene at different temperatures. First, we simulate only the graphene layers without the presence of any gaseous medium, which is presented in Figure 3a. Elevating the temperature of any molecular system increases the thermal vibration of the particles in that system. Ganz et al.⁴⁰ performs ab initio MD simulations to find out the melting point of graphene and reports that the phase change temperature starts at 4000 K, and graphene appears to be in a quasi-2D liquid state at 4500 K. The other limitation is that bilayer graphene is not stable at $T > 3000$ K using the AIREBO potential. In this regard, three different temperatures are tested starting from 300 to 2000 K to investigate the inherent mechanism. It was observed that the fluctuation of the top-layer graphene with respect to the bottom layer increases in both cases, with and without helium.

However, carbon atoms in graphene absorb more thermal energy while it absorbs energy from helium gas. As shown in Figure 3b, the relative RMSD of top-layer graphene is approximately 2 times higher in the presence of helium than without gas at 2000 K. This deviation of top-layer graphene in the presence of gas particles could help weaken the van der Waals bond between graphene layers because the gas particles might find a way to ingress between layers.

In the compressible flow exfoliation (CFE) process, the velocity of gas reaches supersonic flow after passing through a converging–diverging channel because of certain pressure differentials (Figure 1). This is based on the theory of the conservation of mass and momentum:⁴¹

$$\frac{dV}{V}[M^2 - 1] = \frac{dA}{A} \quad (3)$$

where $\frac{dV}{V}$ is the fluids relative change in instantaneous velocity, $\frac{dA}{A}$ is the relative change in the flow channel's instantaneous area, and M is the fluid's Mach number, defined as its velocity V relative to the velocity of the speed of sound, V_s . The flow velocity of gas can achieve even higher velocities than Mach 1, depending on the design of the diverging portion. Because we are more concerned about the area after the throat, where

these high Mach numbers are experienced (supersonic flow), our focus then is to replicate such experimental conditions in our MD simulations. We imparted different flow velocities to helium particles ranging from 1000 to 2000 ms^{-1} . Figure 4a shows the temperature rise of the graphene bilayer with the increase in flow velocities of helium gas. When gas particles collide with carbon atoms, the kinetic energy of the gas flux transfer to the graphene bilayers as thermal energy, which increase its temperature. Therefore, higher flow velocity means more energy transfer and more thermal fluctuations. Regarding the limitation of the AIREBO potential, the results are only valid for $T < 3000$ K, so in this section, we focused on the results in this regime of temperature. As it is shown in Figure 4a, not only graphene reaches 3000 K faster in the higher speed of helium gas, but the RMSD of top-layer graphene, Figure 4b, is higher for $V = 2000 \text{ ms}^{-1}$ ($\text{RMSD} = 16.1 \pm 0.2 \text{ \AA}$) than lower velocities ($\text{RMSD} = 13.2 \pm 0.2$ and $12.0 \pm 0.3 \text{ \AA}$ for $V = 1500$ and 1000 ms^{-1} , respectively). By taking a look at the relation of temperature and RMSD (Figure 4c) at constant velocities, we found that by increasing the velocity, the deviation of graphene layers would reach to a higher amount at lower time, which is the favorable condition in the CFE process. We also observed some bond breaking for all the three velocities as the RMSD exceeds $\sim 10 \text{ \AA}$.

The energy that transferred from gas particles to graphene sheets is demonstrated in Figure 5. The figure shows that by

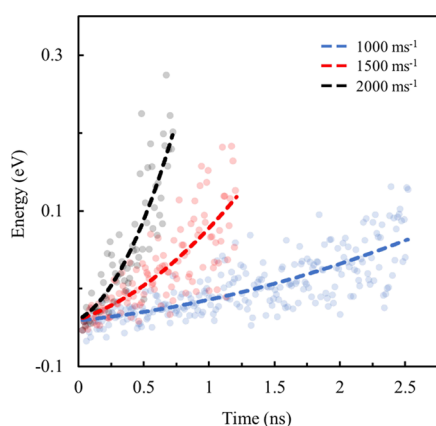


Figure 5. Energy transferred from helium gas to bilayer graphene at different helium velocities ranging from 1000 to 2000 ms^{-1} .

increasing the velocity of helium, the higher amount of energy would be transferred to graphene sheets in less time. The collision of gas particles and graphene sheets at higher gas velocities could raise the kinetic energy of the gas flux transfer to the graphene bilayers as thermal energy, which consequently increases the temperature of the graphene sheet.

Analyzing the structural mechanics of an exfoliating layer of graphene can be performed by calculating the parallel or sliding distance D_{xy} and vertical distance D_z between the centroids of graphene nanosheets during the production simulation. Figure 6 manifests the normalized distribution of sliding and vertical separation distance between graphene layers at three different flow velocities of helium gas, and Table 2 exhibits their corresponding mean values with standard deviation. The mean D_{xy} at 100 ms^{-1} is $0.17 \pm 0.93 \text{ \AA}$, which is increased with a higher standard deviation to $1.47 \pm 2.22 \text{ \AA}$ at 2000 ms^{-1} of helium. The distribution of D_z also exhibits the same trend with the increase in flow velocity, as shown in

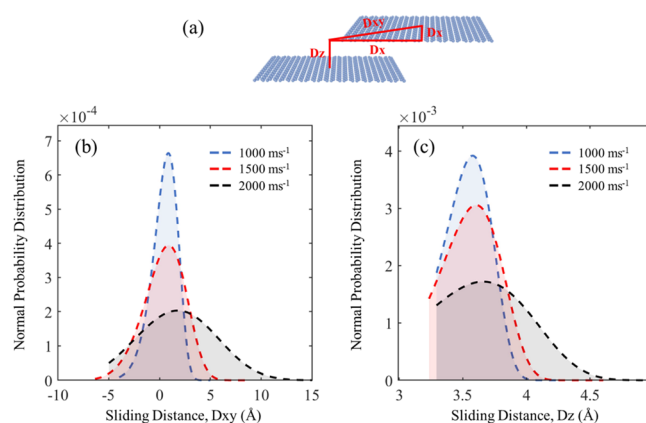


Figure 6. Structural properties of the graphene bilayer at 1000–2000 ms^{-1} velocities of helium gas (a) definition of the sliding and vertical direction of exfoliation. Normalized probability distribution of (b) sliding distance D_{xy} (c) vertical distance D_z of the graphene bilayer.

Figure 6c. At 1000 ms^{-1} , the average D_z is $3.56 \pm 0.02 \text{ \AA}$, which is very close to the equilibrium separation distance between graphene layers. Because of the interlayer vdW interactions, a low flow velocity cannot provide sufficient kinetic energy to overcome diffusion energy barriers. Increasing the flow velocity enables the graphene layers to absorb more energy from the gas, and the average D_z reaches $3.66 \pm 0.26 \text{ \AA}$ at 2000 ms^{-1} . Here, the thermal agitation of carbon atoms in graphene escalates with the ramp of flow velocity, which is the reason for higher sliding and vertical separation between graphene nanosheets. Similar results were also reported during graphene exfoliation by shear-assisted sCO_2 using a rotating mixer.¹⁹ Moreover, the relative in-plane (D_{xy}) fluctuations of graphene layers are much larger compared to those out-of-plane (D_z), as shown by their respective normalized probability distributions in Figure 6. In our study, we have incorporated one-dimensional supersonic flow of gas along the parallel direction—analogue to our CFE experiment—and found that the displacement of graphene layers along the vertical axis is impeded. Thus, we can conclude that sliding is a more likely direction of exfoliation of graphene than the vertical direction during the CFE process.

The density and the flow velocity of fluid rely on the pressure differential of the convergent–divergent section of the CD nozzle. If the upstream supply pressure is not notably above ambient at all times, the CD nozzle will only choke at the throat. In this case, it will act as a Venturi tube, and no supersonic flow will be achieved. Again, the density of fluid particles is interlinked with the pressure at that point according to the ideal gas law. Therefore, the upstream pressure of gas plays a significant role in the exfoliation of graphene nanosheets during the CFE process. In our study, we have used two different upstream stagnant pressures of gas—200 and 800 psi to investigate the effect of pressure. From the temperature plot of the exfoliating layer of graphene, it is evident that it would take more than 4 times longer to reach $T = 3000 \text{ K}$ at low pressure compared to the high pressure, as shown in Figure 7a. Because of the low density at 200 psi, there are fewer collision between carbon atoms and helium gas particles. As a result, graphene layers cannot absorb sufficient kinetic energy to overcome the energy barriers of interlayer vdW interactions. Figure 7b shows the energy that absorbed by graphene from helium gas at 200 and 800 psi pressures. The

Table 2. Mean sliding (D_{xy}) and vertical (D_z) distance between two graphene nanosheets at three different flow velocities of helium gas with standard deviation.

flow velocity (ms^{-1})	mean D_{xy} (Å)	mean D_z (Å)
1000	1.47 ± 2.22	3.66 ± 0.26
1500	0.17 ± 0.93	3.56 ± 0.02
2000	0.46 ± 1.01	3.60 ± 0.06

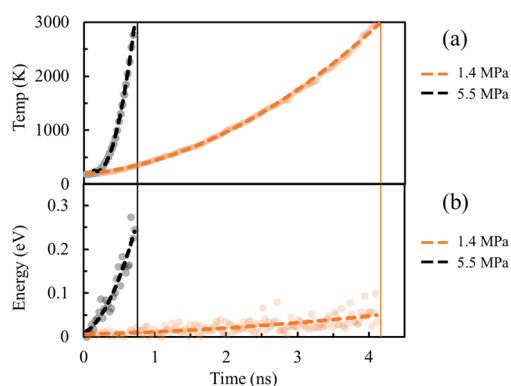


Figure 7. Effect of the supply pressure of gas on graphene exfoliation (a) temperature of graphene (b) the energy transferred from helium gas to the bilayer graphene at different helium pressures at 200 and 800 psi.

graphene at lower pressure (200 psi) only $E \sim 0.04$ eV at $T = 3000$ K while this number is $E \sim 0.25$ for the higher pressure. Increasing the pressure from 200 to 800 psi, the density of helium gas increases (from ~ 0.78 to ~ 3.08 Kg m^{-3}), which intensifies the absorbed energy by graphene. Therefore, a high supply pressure of gas is more suitable for the exfoliation of 2D nanomaterials using the CFE process. Moreover, the rising temperature (kinetic energy) mostly depends on the relative atomic mass of monoatomic gas and slightly change by atmospheric pressure. However, as shown by Tao et al.⁴² in their study of aerodynamic heating of hypersonic re-entry

materials, increasing pressure would raise the collision of the gas atoms with graphene and thus excite phonons with the specific frequency, which in our case, would make major contribution to the heating of the graphene.

The final series of MD simulations investigated the effect of two different monoatomic gases on the exfoliation of graphene nanosheets, and the results are summarized in Figure 8. Argon has an atomic mass of 39.948u, which is significantly higher than helium (4.003u), and is considered here for comparison. Monoatomic gases are chosen in this study for the simplicity of modeling the MD simulation. Although argon has a higher density than helium in the same experimental conditions, it has much slower speeds because of its high atomic mass. The Mach profile (velocity profile) in convergence–divergence nozzles is the same for ideal gases. In the same nozzle, the speed of argon gas is lower than that of helium because the speed of sound is lower in heavier ideal gases (V_{sound} : Ar (300 K) = 323 m/s, He (300 K) = 965 m/s). Based on this fact, here, we used a lower speed for Ar than He in our MD simulations ($V_{\text{Ar}}/V_{\text{He}} = \sim 1/3$). The maximum deviation of the exfoliating layer of graphene is only ~ 3 Å in argon, whereas it reaches up to more than 16 Å for the case of helium, as shown in Figure 8a. Therefore, the graphene layers cannot absorb sufficient kinetic energy from the argon gas flux to cause temperature increase and bond breaking, leading to exfoliation. To obtain the same level of fluctuations from gases with high atomic mass, the upstream pressure or the cross-sectional area of divergent section must be considerably increased. However, continuous operations at high pressure may bring serious safety concerns and incur more cost to the process. The

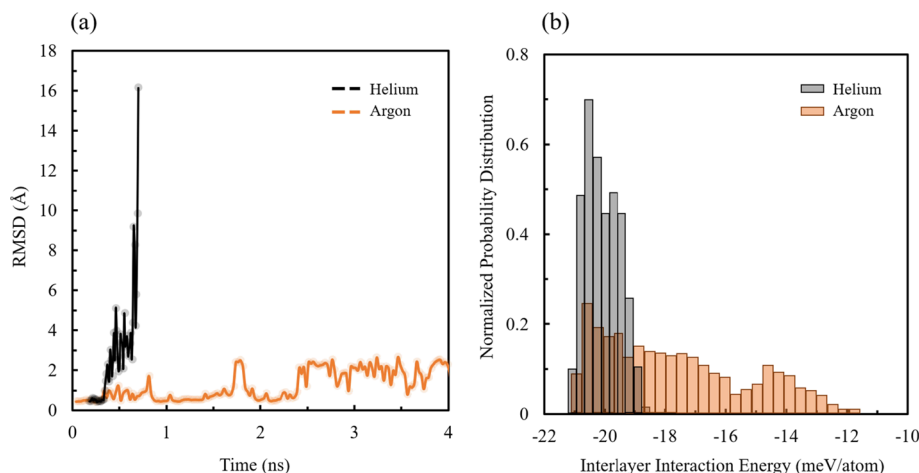


Figure 8. Simulated exfoliation phenomena of the graphene bilayer in two different gas medium (a) RMSD of the exfoliating layer of graphene and (b) normalized interlayer interaction energy between graphene layers in helium and argon gases.

interlayer interaction energy of graphene (E_{gg}) in the existence of argon and helium and its distribution is shown in Figure 7b. The E_{gg} in argon and helium exhibits distinctively different distributions. The average of the absolute value of E_{gg} in argon is 19.57 ± 0.56 meV/atom, which is higher than that of helium (17.15 ± 2.45 meV/atom), which represent that the collision between helium and graphene can better weaken the interaction energy between graphene layers than argon and graphene. Based on seven different exchange-correlation ab initio functional theory, the absolute binding energy between graphene layers⁴³ was calculated as 23.67 meV/atom for AB stacked graphene in vacuum at an equilibrium interlayer distance (3.334 Å), and this energy decreases with the increase in the separation distance. In our simulations, the distance between graphene layers in helium gas is considerably higher than that for argon because of higher thermal fluctuations. Consequently, this difference of interlayer separation induced lower binding energy for the case of helium.

CONCLUSIONS

To comprehend the effect of gas particles on the exfoliation properties of graphite during the CFE process, we performed classical molecular dynamics-based study of a system composed of two graphene nanosheets placed along the direction of the supersonic flow of gas. A versatile hybrid formulation of AIREBO-M and LJ potential functions is used to describe the interactions of a graphene–noble gas system. The findings of this study provide some useful information that may allow to maximize process yield through the design of experiments and increase the versatility of the overall process. Our results show that increasing the temperature of the system intensifies the fluctuations of the graphene layers. Analysis of flow simulation indicates that the relative fluctuations and diffusivity of the exfoliating layer of graphene is higher in the high flow velocity of gas particles. The low velocity of gas cannot provide sufficient kinetic energy to the carbon atoms to overcome the diffusion energy barriers between graphene nanosheets because of interlayer interaction energy. Our calculations indicate that exfoliation is favored in sliding rather than in the vertical direction because of one-dimensional in-plane supersonic flow of gas. The density and flow velocity of gas are interlinked with upstream pressure and the delamination of graphene is easier at high pressures because of an increase in the number of collisions between gas particles and carbon atoms. The selection of gas media also plays a significant role in the CFE process. Our computational study shows that a lighter gas (helium) has superiority over heavier gas (argon) in exfoliating graphene because of its higher flow velocity at the same primary conditions. The larger fluctuations of graphene sheets in helium gas also induce more interlayer interactions energy, which is -2.42 ± 1.89 meV/atom higher than that in argon. We, therefore, propose that a lighter gas media with high upstream pressure will provide the best exfoliation of graphene through the CFE process. The result of this study on the effect of abovementioned parameters, such as temperature, pressure, and the type of gas, would be also applicable, qualitatively, on many-layer graphene. As we discussed in the study, increasing the temperature facilitates the separation in bilayer graphene by transferring energy to it; likewise, many-layer graphene would also absorb more energy on higher temperature, although this energy might be insufficient to overcome every layer of van der Waals force. We hope that our study provides necessary insights at a

molecular level, which can be used as a guideline for the exfoliation of not only graphene but also any 2D layered materials using compressible gases.

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Notes

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